Exp 10

Create and Explore a Data Frame

Aim:

To create and explore a data frame named `exam_data` with name, score, attempts, and qualifying fields. To extract data, add a row and column, sort the data, and save it to a file.

Procedure:

1. Create a data frame with the required fields.
2. Display the data frame and extract selected values.
3. Add a new row and a new column.
4. Sort the data frame by score.
5. Save the data frame as a CSV file.

Code:

```r
exam_data <- data.frame(
  name = c("Arun", "Bala", "Cathy", "David"),
  score = c(85, 72, 90, 65),
  attempts = c(1, 2, 1, 3),
  qualifying = c(TRUE, TRUE, TRUE, FALSE)
)

print(exam_data)

print(exam_data$name)
print(exam_data[1, ])
print(exam_data$score)

new_row <- data.frame(
  name = "Eva",
  score = 88,
  attempts = 1,
  qualifying = TRUE
)

exam_data <- rbind(exam_data, new_row)
exam_data$grade <- c("A", "B", "A", "C", "A")

print(exam_data)

sorted_data <-
exam_data[order(exam_data$score), ]
print(sorted_data)

write.csv(exam_data, "exam_data.csv",
row.names = FALSE)
print("Data saved successfully")
```

Output:

```
1  Arun    85        1       TRUE
2  Bala    72        2       TRUE
3 Cathy    90        1       TRUE
4 David    65        3      FALSE

[1] "Arun" "Bala" "Cathy" "David"

   name score attempts qualifying
1  Arun    85        1       TRUE

[1] 85 72 90 65
```

```
   name score attempts qualifying grade
1  Arun    85        1       TRUE     A
2  Bala    72        2       TRUE     B
3 Cathy    90        1       TRUE     A
4 David    65        3      FALSE     C
5   Eva    88        1       TRUE     A

   name score attempts qualifying grade
4 David    65        3      FALSE     C
2  Bala    72        2       TRUE     B
1  Arun    85        1       TRUE     A
5   Eva    88        1       TRUE     A
3 Cathy    90        1       TRUE     A

[1] "Data saved successfully"
```

Result:

Thus, the `exam_data` data frame was created, manipulated, sorted, and saved successfully as a CSV file.